TX-OMICS REVISION

# Seven gene sets.
# Seven clear definitions.

Family 1 captures four primary evidence categories. Family 2 contains three CSW-derived GO term groups, with FC0.5 and FC1 provenance shown separately.

**4**
primary sets

**3**
GO-derived sets

**2**
thresholds

**7**
stocker CSVs

# Two families, seven non-interchangeable sets

Each set now receives one slide; overlap is reported separately within each family.

**FAMILY 1 · PRIMARY EVIDENCE**

| | | | |
|---|---|---|---|
| 01 | 6 | **Mechanistic** | Consistent/correlated + published homeostatic-consistent effect |
| 02 | 20 | **Homeostatic genes** | Opposing history versus rebound correlations |
| 03 | 97 | **CSW 4+ genes** | Consistent in four or more of seven datasets |
| 04 | 4 | **HLH genes** | Potential upstream regulators |

**FAMILY 2 · CSW-DERIVED GO**

| | | | |
|---|---|---|---|
| 05 | 99 | **Ribosomal** | FC0.5 99 · FC1 12 |
| 06 | 167 | **Mitochondrial** | FC0.5 167 · FC1 23 |
| 07 | 5 | **Immune** | FC0.5 0 · FC1 5 |

**FC0.5 and FC1 counts are source-provenance counts. A gene may appear at both thresholds.**

# Mechanistic · 6 genes

Consistent or correlated sleep/wake genes with published effects compatible with homeostatic feedback.

6

**paper-highlighted genes**

Inclusion requires transcriptomic evidence plus published sleep/wake function.

- Consistent across sleep/wake conditions or correlated with sleep/wake history
- Published perturbation changes sleep/wake
- Effect direction fits a potential homeostatic loop

**unc79**
NALCN complex

**SIFa**
neuropeptide

**rumpel**
glial transporter

**AstA-R2**
neuropeptide receptor

**Trhn**
serotonin synthesis

**RpL23**
ribosome / translation

**These six are not the 20 opposing history-rebound correlation genes.**

# Homeostatic genes · 20 genes

Genes correlated with pre-sampling sleep/wake history and post-sampling rebound in the opposite direction.

**15**

**History+ / Rebound−**

Positive correlation with prior sleep; negative correlation with rebound.

**5**

**History− / Rebound+**

Negative correlation with prior sleep; positive correlation with rebound.

15 · HISTORY+ / REBOUND−

**AstA-R2 · bumpel · CG33080 · CG41378 · CG7460**
**CG7601 · Chrac-14 · Cyp28c1 · dpr21 · kumpel**
**Obp44a · Sk1 · Ssk · Trhn · VGlut2**

5 · HISTORY− / REBOUND+

**CG42323 · CG7079 · CG9313 · CG9377 · Gtpbp1**

# CSW 4+ genes · 97 genes

Consistent sleep/wake genes evident across four or more of the seven datasets.

**97**

**FC0.5 wake genes**

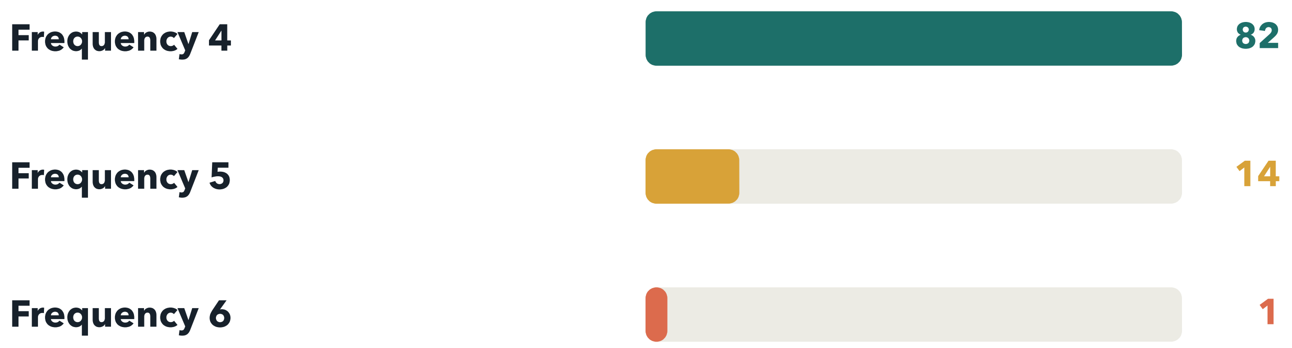
The category rule is recurrence across datasets—not GO enrichment.

**7**

**also qualify at FC1**

All seven are already contained within the 97-gene FC0.5 set.

## Recurrence across seven datasets




**No sleep-upregulated gene met the ≥4-of-7 recurrence criterion.**

# HLH genes · 4 genes

Four bHLH factors identified as potential upstream regulators of wake-induced gene expression.

REGULATORY LOGIC

## CAGCTG

E-box enriched upstream of wake-induced genes

**HOMER motif enrichment → candidate direct regulators**

Results · Table S1

### bigmax

FBgn0039509

### HLH3B

FBgn0011276

### E(spl)m3-HLH

FBgn0002609

### E(spl)mbeta-HLH

FBgn0002733

**This set is regulatory evidence—not frequency or correlation evidence.**

# Ribosomal / translation · 99 genes

GO-report assignment is shown here; experiment-level FC0.5 and FC1 regulation is shown on slide 12.

**GO ASSIGNMENT SOURCE**

**99**

unique genes in this set

| | | |
|---|---|---|
| FC0.5 GO | 99 | genes assigned |
| FC1 GO | 12 | genes assigned |
| BOTH | 12 | assigned by both reports |

**99 FC0.5 + 12 FC1 − 12 both = 99 unique**

## Most frequent approved terms

| | |
|---|---|
| Ribosome | 93 |
| Structural constituent of ribosome | 92 |
| Translation | 77 |
| Cytoplasmic translation | 65 |
| Cytosolic ribosome | 58 |

87 FC0.5-only · 0 FC1-only · 12 at both thresholds

# Mitochondrial / metabolism · 167 genes

GO-report assignment is shown here; experiment-level FC0.5 and FC1 regulation is shown on slide 13.

**GO ASSIGNMENT SOURCE**

**167**

unique genes in this set

| | | |
|---|---|---|
| FC0.5 GO | **167** | genes assigned |
| FC1 GO | **23** | genes assigned |
| BOTH | **23** | assigned by both reports |

**167 FC0.5 + 23 FC1 − 23 both = 167 unique**

## Most frequent approved terms

| | |
|---|---|
| **Mitochondrion** | 121 |
| **Mitochondrial inner membrane** | 72 |
| **Oxidative phosphorylation** | 46 |
| **Mitochondrial translation** | 30 |
| **Mitochondrial matrix** | 24 |

**144 FC0.5-only · 0 FC1-only · 23 at both thresholds**

# Immune · 5 genes

The set was assigned from FC1 GO enrichment; all five genes also meet FC0.5 regulation criteria.

GO ASSIGNMENT SOURCE

**5**

unique genes in this set

| | | |
|---|---|---|
| FC0.5 GO | 0 | genes assigned |
| FC1 GO | 5 | genes assigned |
| BOTH | 0 | assigned by both reports |

**0 FC0.5 + 5 FC1 − 0 both = 5 unique**

## Five FC1-supported genes

- BomBc2
- BomS1
- BomS2
- IM14
- Thor

### Approved GO terms · genes per term

| | |
|---|---|
| Defense response | 4 |
| Antibacterial humoral response | 4 |
| Response to bacterium | 4 |

Genes may occur in more than one approved term.

**0 FC0.5-only · 5 FC1-only · 0 at both**

# Only three genes cross primary-set boundaries

Overlap is calculated among Mechanistic, Homeostatic, CSW 4+, and HLH only.

## Observed shared genes

| | | |
|---|---|---|
| **Mechanistic × Homeostatic** | 2 | **Trhn · AstA-R2** |
| **Mechanistic × CSW 4+** | 1 | **RpL23** |
| **All other pairings** | 0 | No shared FBgns |

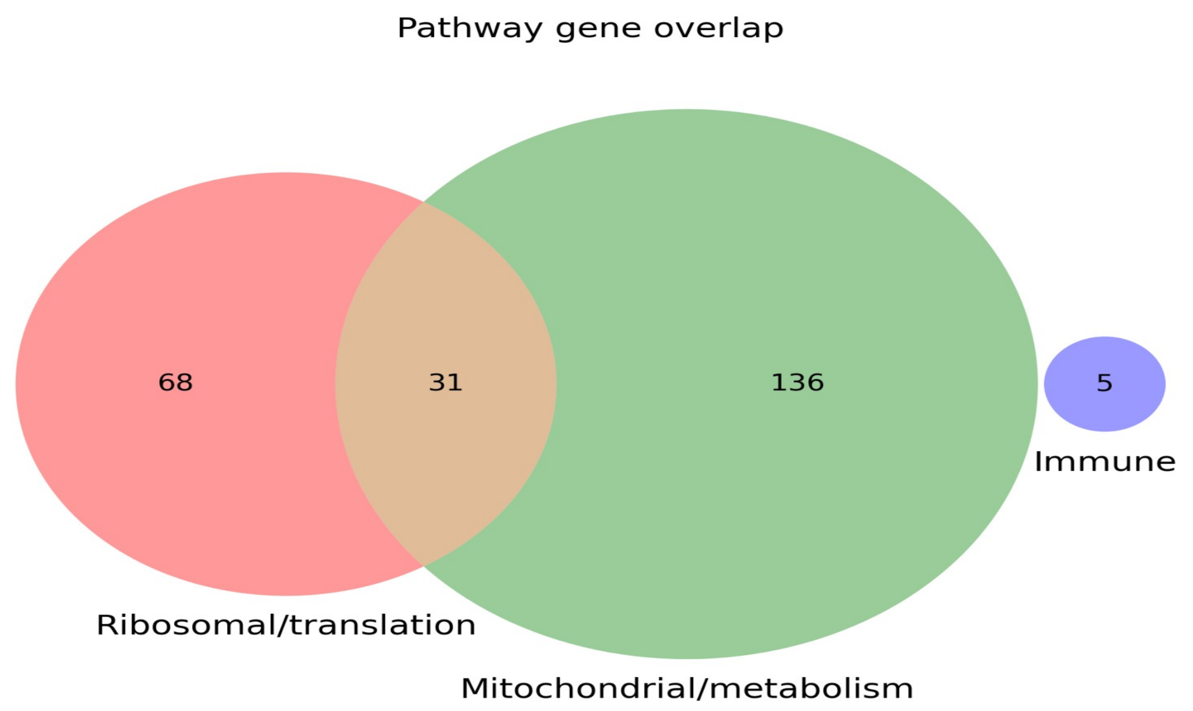
## Family 1 union

| | |
|---|---|
| 124 | unique FBgns |
| 121 | in exactly one set |
| 3 | in exactly two sets |
| 0 | in three or four sets |

**HLH genes are fully disjoint.**

# The GO-derived union contains 240 unique genes

99 ribosomal + 167 mitochondrial + 5 immune − 31 shared ribosomal/mitochondrial genes = 240 unique FBgns.




**240 = union of Ribosomal, Mitochondrial and Immune genes; each FBgn counted once.**

## GO-assignment provenance across the 240-gene union

**203** FC0.5 only

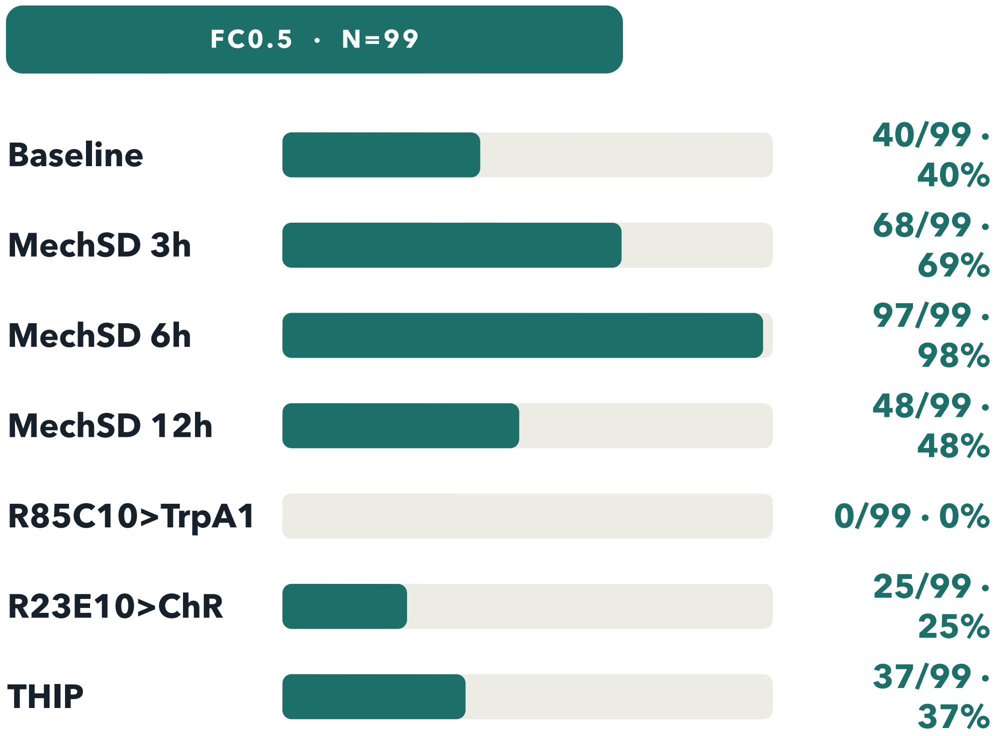
**32** both thresholds

**5** FC1 only

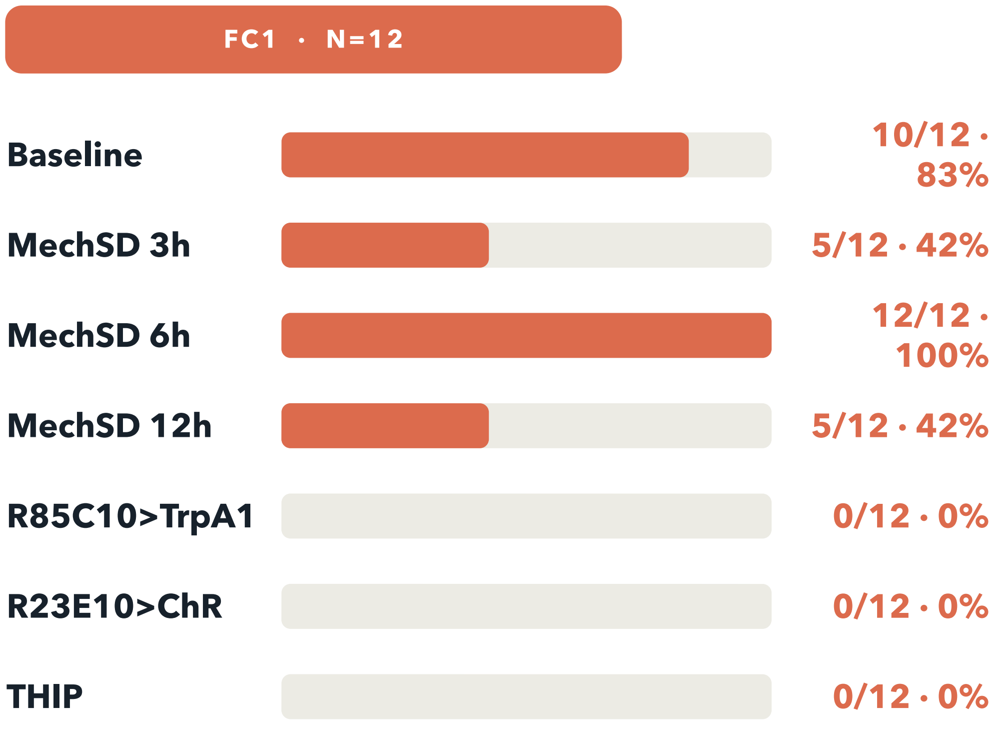
Only the five Immune genes are FC1-exclusive.

# Ribosomal / translation · experiment-level regulation

Bars show the share of category genes with logFC > threshold and q < 0.05 in each experiment; genes may appear in multiple bars.




n = category genes meeting this threshold in at least one experiment

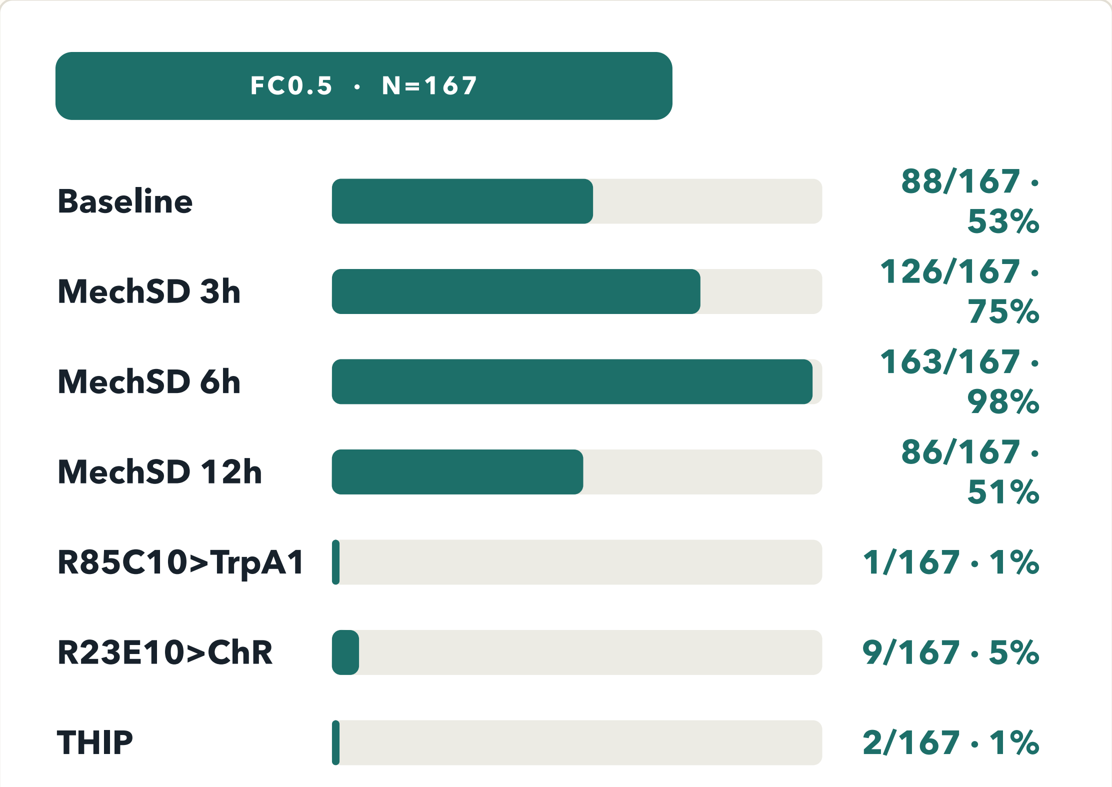



n = category genes meeting this threshold in at least one experiment

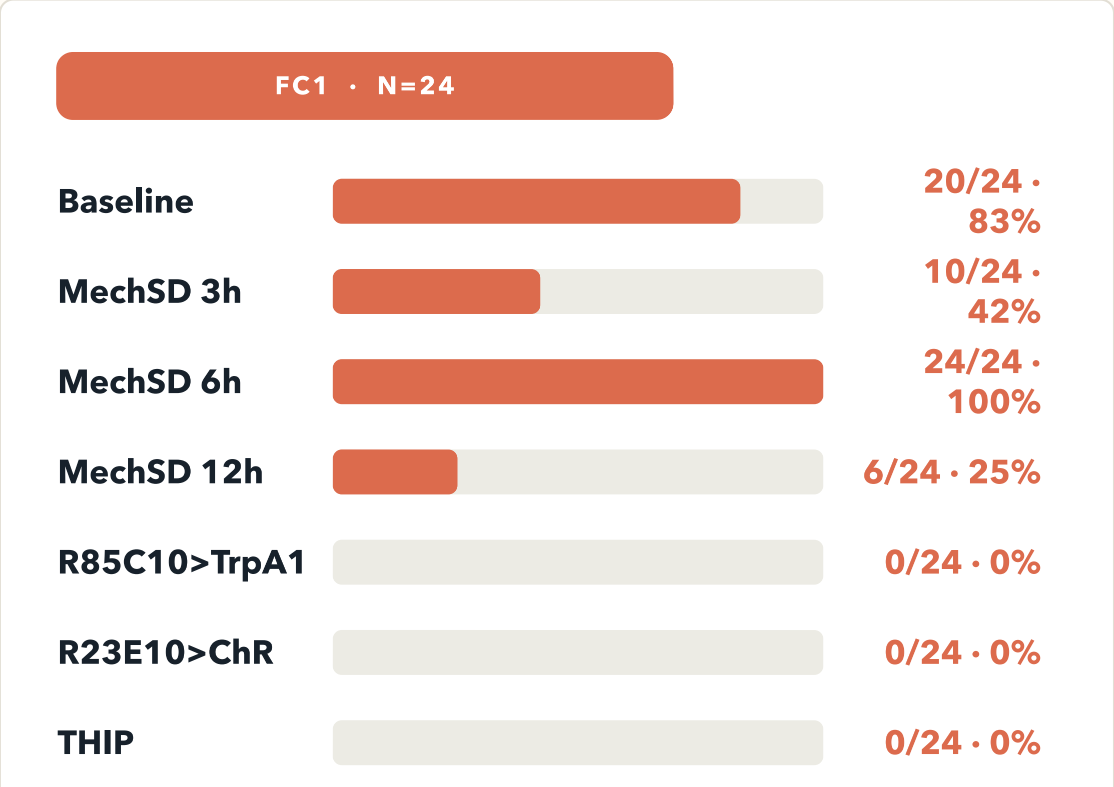
**MechSD6 regulates nearly all ribosomal genes at both thresholds; FC1 signal is concentrated in Baseline and mechanical deprivation.**

# Mitochondrial / metabolism · experiment-level regulation

Bars show the share of category genes with logFC > threshold and q < 0.05 in each experiment; genes may appear in multiple bars.




n = category genes meeting this threshold in at least one experiment




n = category genes meeting this threshold in at least one experiment

**MechSD6 is the dominant mitochondrial response at both thresholds; FC1 hits are absent from R85C10, ChR and THIP.**

# Immune · experiment-level regulation

Bars show the share of category genes with logFC > threshold and q < 0.05 in each experiment; genes may appear in multiple bars.

**FC0.5 · N=5**

| | |
|---|---|
| Baseline | 1/5 · 20% |
| MechSD 3h | 4/5 · 80% |
| MechSD 6h | 3/5 · 60% |
| MechSD 12h | 4/5 · 80% |
| R85C10>TrpA1 | 0/5 · 0% |
| R23E10>ChR | 0/5 · 0% |
| THIP | 0/5 · 0% |

n = category genes meeting this threshold in at least one experiment

**FC1 · N=5**

| | |
|---|---|
| Baseline | 1/5 · 20% |
| MechSD 3h | 4/5 · 80% |
| MechSD 6h | 3/5 · 60% |
| MechSD 12h | 4/5 · 80% |
| R85C10>TrpA1 | 0/5 · 0% |
| R23E10>ChR | 0/5 · 0% |
| THIP | 0/5 · 0% |

n = category genes meeting this threshold in at least one experiment

**All five immune genes meet both regulation thresholds; MechSD3 and MechSD12 regulate four of five genes.**